

Request for a Cost-Effective AWS EBS Storage Extension

For fine-tuning the production-grade streaming ASR model on the 25-hour in-house debt collection dataset

Executive summary

I am requesting a small extension of the current root EBS volume from about 200 GB to about 300 GB (an additional 100 GB). This is a low-cost, operationally necessary change to safely complete fine-tuning for our 600M-parameter Indic-Conformer model without disk-related interruptions. The request is for storage only, not for a larger GPU or server upgrade.

Current state

Item	Current value
Training task	Fine-tuning the production-grade streaming ASR model on a 25-hour in-house debt collection dataset
Model / scale	Indic-Conformer model with roughly 600 million parameters
Current storage	/dev/root = 193 GB total, 170 GB used, 24 GB free, 88% utilized after cleanup
Observed risk	Disk utilization previously reached 99%, which halted the training pipeline
Current GPU	Tesla T4 (16 GB VRAM), retained intentionally as the cheaper compute option
Requested change	Extend only the root EBS volume by 100 GB to reach about 300 GB total

Why this is a blocker right now

Model snapshot generation: Training must save checkpoints regularly so progress is not lost during interruptions and so the best-performing model version can be selected later. Each checkpoint for the 600M model is expected to take roughly 7.5 GB, so even a minimal retention policy quickly consumes the remaining free space.

Audio pre-processing overhead: During feature extraction, the training pipeline temporarily caches acoustic representations such as spectrogram-like features. This can briefly inflate the on-disk footprint beyond the raw .wav dataset size and reduce the free-space buffer much faster than the base dataset alone suggests.

Operational safety margin: Training runs also need space for logs, temporary files, package caches, and ordinary OS activity. Once the root disk gets close to full, the risk of failed writes, stalled processes, or interrupted training increases sharply.

Why this is the most cost-effective solution

Point	Business rationale
Small monthly cost	AWS gp3 storage is priced at about \$0.08 per GB-month in AWS pricing examples, so an additional 100 GB is approximately \$8 per month before taxes and region-specific differences.
Avoids wasted compute	A failed 15+ hour training run on the current GPU is more expensive than a small storage increase because the compute time is already paid for but produces no usable training result if the disk fills up mid-run.
No massive infrastructure request	This request does not ask for an A10G, A100, or a broader instance migration. It preserves the existing lower-cost T4 setup and solves the immediate blocker with the cheapest practical infrastructure change.
Fast operational path	Amazon EBS supports online volume expansion on supported instances; after the volume is enlarged, the partition and file system can be extended so the training environment can resume with minimal disruption.

What I am not requesting

I am not asking for a major GPU upgrade right now. Although the current Tesla T4 is slower for fine-tuning, I am intentionally continuing with this cheaper setup to keep compute costs controlled. The storage extension is the narrowest and most economical change required to make the existing training setup reliable. The A10G comparison included later is only a planning reference to show the speed tradeoff if iteration time becomes a bottleneck in the future.

Requested action

Please extend the current root EBS volume by 100 GB, from about 193 GB to about 300 GB total, so that I can safely resume and complete the training run without disk-capacity failures.

Manager takeaway

This is a small storage-only request that removes an immediate training blocker, protects already-paid compute time, and avoids a much more expensive GPU upgrade.

Illustrative training-time comparison (planning only)

To estimate fine-tuning duration, we measure **time per epoch** — where one epoch means the model has seen the full 25-hour dataset once. The figures below are rough planning estimates, not guaranteed benchmarks; actual timings can vary with audio-length distribution, feature extraction settings, mixed precision, checkpoint cadence, and dataloader throughput.

Setup	Batch	Time / epoch	10 epochs	Practical note
Tesla T4 (16 GB)	1	~3.0 h	~30 h	Safest setup
Tesla T4 (16 GB)	2	~2.0–2.5 h	~20–25 h	Practical current target
Tesla T4 (16 GB)	4	Likely OOM	—	May exceed VRAM
NVIDIA A10G (24 GB)	2	~45–60 min	~7.5–10 h	Faster even at same batch
NVIDIA A10G (24 GB)	4	~35–45 min	~5.8–7.5 h	Good balance
NVIDIA A10G (24 GB)	8	~20–30 min	~3.3–5 h	High GPU utilization
NVIDIA A10G (24 GB)	16	Possible OOM	—	Depends on audio lengths

Interpretation: This comparison is included only to show why a future A10G move would reduce iteration time substantially. It is **not** part of the current request. The immediate and cost-effective ask remains a 100 GB EBS extension so the existing T4-based run can complete reliably.

Bottom line
The storage extension is the smallest spend that removes the immediate blocker. The GPU comparison is only a reference to show that I am deliberately avoiding a more expensive compute upgrade right now.